

UNIVERSITY OF WAH(UOW)
DEPARTMENT OF COMPUTER
SCIENCE



Lab Task # 02

COURSE: Data Structure & Algorithm Lab_201L

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Task # 01

```
#include <iostream>
#include <cstdlib>
#include <ctime>

using namespace std;

int main() {
    srand(time(0));
    cout << "10 random numbers between 1 and 100:" << endl;
    for (int i = 0; i < 10; i++) {
        int num = rand() % 100 + 1;
        cout << num << " ";
    }
    cout << endl;
    return 0;
}
```

Output

```
10 random numbers between 1 and 100:
69 53 31 25 65 96 87 56 21 78
```

Task # 02

```
struct Student {
    string name;
    int rollNo;
    float marks;
};

int main() {
    Student s[3];
    cout << "Enter details of 3 students: "<<endl;
    for (int i = 0; i < 3; i++) {
        cout << "\nStudent " << i + 1 << ":\n";
        cout << "Enter name: ";
        cin >> s[i].name;
        cout << "Enter roll number: ";
        cin >> s[i].rollNo;
        cout << "Enter marks: ";
        cin >> s[i].marks;
    }
    cout << "\nStudent Details"<<endl;
    for (int i = 0; i < 3; i++) {
        cout << "Name: " << s[i].name << ", Roll No: " << s[i].rollNo
            << ", Marks: " << s[i].marks << endl;
    }
    return 0;
}
```

OutPut

```
Student 1:  
Enter name: Talhs  
Enter roll number: 222  
Enter marks: 100
```

```
Student 2:  
Enter name: waqa  
Enter roll number: 111  
Enter marks: 433
```

```
Student 3:  
Student 2:  
Enter name: waqa  
Enter roll number: 111  
Enter marks: 433
```

```
Student Details  
Name: Talhs, Roll No: 222, Marks: 100  
Name: waqa, Roll No: 111, Marks: 433  
Name: ree, Roll No: 333, Marks: 221
```

Task # 03

```
int main() {  
    double num;  
    cout << "Enter a number: ";  
    cin >> num;  
    cout << "\nResults using math functions:\n";  
    cout << "Square root: " << sqrt(num) << endl;  
    cout << "Power (num^2): " << pow(num, 2) << endl;  
}
```

Output

```
Enter a number: 22  
  
Results using math functions:  
Square root: 4.69042  
Power (num^2): 484
```

Task # 04

```

int main() {
    float temp;
    int weather;
    cout << "Enter current temperature: ";
    cin >> temp;
    cout << "Select weather condition: "<<endl;;
    cout << "1. Sunny"<<endl;
    cout << "2. Rainy"<<endl;
    cout << "3. Cloudy"<<endl;
    cout << "Enter your choice (1-3): ";
    cin >> weather;
    cout << "\nAdvice: ";
    switch (weather) {

```

```

        case 1:
            if (temp > 35)
                cout << "Too hot stay indoors" << endl;
            else
                cout << "Nice sunny day" << endl;
            break;
        case 2:
            cout << "Carry an umbrella!" << endl;
            break;
        case 3:
            if (temp < 15)
                cout << "Cold and cloudy, wear warm clothes." << endl;
            else
                cout << "Cloudy but comfortable weather." << endl;
            break;
        default:
            cout << "Invalid weather condition selected!" << endl;
    }
}

```

Output

```

Enter current temperature: 44
Select weather condition:
1. Sunny
2. Rainy
3. Cloudy
Enter your choice (1-3): 1

Advice: Too hot stay indoors

```

Task # 05

```

#include <iostream>
using namespace std;
bool isVowel(char ch) {
    ch = tolower(ch);
    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
        return true;
    else
        return false;
}

int main() {
    char c;
    cout << "Enter a character: ";
    cin >> c;

    if (isVowel(c))
        cout << c << " is a vowel." << endl;
    else
        cout << c << " is not a vowel." << endl;
    return 0;
}

```

Output

```

Enter a character: e
e is a vowel.

```

Task # 06

```

#include <iostream>
using namespace std;
long long factorial(int n) {
    long long fact = 1;
    for (int i = 1; i <= n; i++) {
        fact = fact * i;
    }
    return fact;
}

int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;
    if (num < 0)
        cout << "Factorial is not defined for negative numbers." << endl;
    else
        cout << "Factorial of " << num << " is: " << factorial(num) << endl;
    return 0;
}

```

Output

```
Enter a number: 10
Factorial of 10 is: 3628800
```
